

a

Full reward · $H(\text{ZOH}) = 255.9$



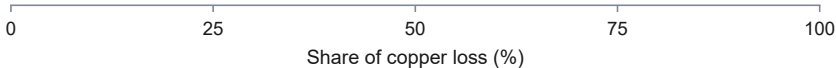
filter-reachable 18.1%

No torque + energy · $H(\text{ZOH}) = 376.3$ (+47%)



filter-reachable 43.3%

Removing the effort reward terms moves 25 percentage points of heat into the band a filter can reach.



DC (posture) 0–5 Hz (gait) 5–25 Hz (in-band jitter) >25 Hz (ZOH harmonics)

b

47% of 18.1% recovered

0.915

54% of 43.3% recovered

0.765

H0 baseline = 1.00

Raw savings differ by nearly 3×; the recovered fraction is stable.

